

KAVIYA V

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CAREER OBJECTIVE

Aspiring AI Engineer skilled in Python, Machine Learning, Deep Learning, and Computer Vision, seeking an entry-level role to build impactful AI solutions and grow as a technology professional.

TECHNICAL SKILLS

Programming: Python, SQL, C
Databases: MySQL
ML & AI: Machine Learning, Deep Learning, Computer Vision, NLP, CNN, LSTM, LLM, RAG
Frameworks & Libraries: TensorFlow, Keras, Scikit-learn, OpenCV, FastAPI, Streamlit, Pandas, NumPy
Tools: Git, GitHub, Jupyter Notebook, Google Colab, VS Code, Power BI

INTERNSHIP

Data Science Intern | Accent TechnoSoft Jun 2025 – Jul 2025

- Built Random Forest classifier achieving 89% accuracy on biodiversity dataset of 10,000+ species records for extinction risk classification.
- Performed data cleaning, feature engineering, and EDA using Pandas and Matplotlib to identify key conservation predictors.
- Selected optimal model based on precision, recall, and F1-score metrics enabling data-driven wildlife conservation insights.

PROJECTS

Sign Language to Text Recognition
Tech Stack: TensorFlow, Keras, OpenCV, Python

- Engineered CNN-based real-time sign language recognition system using 3,000+ preprocessed gesture images with augmentation techniques (rotation, flipping, normalization).
- Achieved 85% classification accuracy; deployed interactive interface using OpenCV for live video inference, enhancing accessibility.

Student Grade Predictor — ML Pipeline
Tech Stack: Scikit-learn, Pandas, NumPy, Python

- Built and compared 3 classification models (Logistic Regression, Decision Tree, Random Forest) to predict student grades across 5 categories on 2,392 student records, achieving 75.4% accuracy after tuning.
- Engineered 6 new features (study efficiency, risk score, support score) that improved feature-target correlation up to 3x; deployed a production-ready Scikit-learn Pipeline with StandardScaler to prevent data leakage.
- Evaluated models using 5-fold cross-validation and F1-score; tuned classification threshold to optimize the precision-recall tradeoff.

Sentiment Analysis — Deep Learning NLP
Tech Stack: TensorFlow, Keras, LSTM, Python

- Developed a Bidirectional LSTM neural network with Embedding and Dropout layers to classify product review sentiment using TensorFlow/Keras.
- Built an end-to-end NLP preprocessing pipeline (text cleaning, tokenization, sequence padding) and prevented data leakage by fitting the tokenizer strictly on training data.
- Diagnosed model overfitting through train/test accuracy gap analysis and documented root-cause findings to guide future regularization strategy.

Student Performance Analyzer — EDA
Tech Stack: Pandas, NumPy, Python

- Performed exploratory data analysis on 2,392 student records using Pandas and NumPy, identifying absenteeism as the strongest performance driver (correlation: -0.919).
- Delivered data-driven insights via correlation and group-by analysis, including at-risk student identification (34% of cohort) and tutoring impact (+0.29 GPA).

EDUCATION

Kangayam Institute of Technology | B.Tech Artificial Intelligence & Data Science 2023 – 2027
CGPA: 8.21/10

CERTIFICATIONS

- Google Analytics Certification — Google
- AI Engineer Certification — Udemy

ACHIEVEMENTS & CO-CURRICULAR ACTIVITIES

- Vice President, ISTE — Organized technical workshops and student development events promoting engineering excellence.
- Smart India Hackathon Participant — Created datasets and developed solutions for Transportation & Logistics domain at national-level hackathon.

LANGUAGES

Tamil (Native), English (Professional Working Proficiency)